

Chapter 10

Applications of Differentiation

Why Should We Study Applications of Differentiation?

Differentiation is not only a mathematical concept; it is a powerful tool used in real life.

This chapter helps us understand:

- How fast a quantity changes
- How businesses calculate profit and cost
- How economists study revenue and production
- How scientists analyze motion and growth
- How engineers optimize machines and structures
- How graphs behave in real-world situations

10.1 Introduction

Applications of differentiation play an important role in Economics, Physics, Engineering, Biology and Data Science.

Using derivatives, we can determine:

- Growth rate
- Average cost
- Marginal cost
- Marginal revenue

- Maximum profit
- Minimum cost
- Rate of motion

Differentiation helps us make better decisions using mathematics.

10.2 Growth Rate

Growth rate measures how rapidly a quantity changes with time.

If:

$$y = f(t)$$

then:

$$\frac{dy}{dt}$$

represents the growth rate.

Growth Rate Formula

$$\text{Growth Rate} = \frac{dy}{dt}$$

Solved Example

Suppose population is given by:

$$P(t) = 1000 + 50t + 2t^2$$

Find the growth rate when:

$$t = 5$$

Solution

Differentiate:

$$\frac{dP}{dt} = 50 + 4t$$

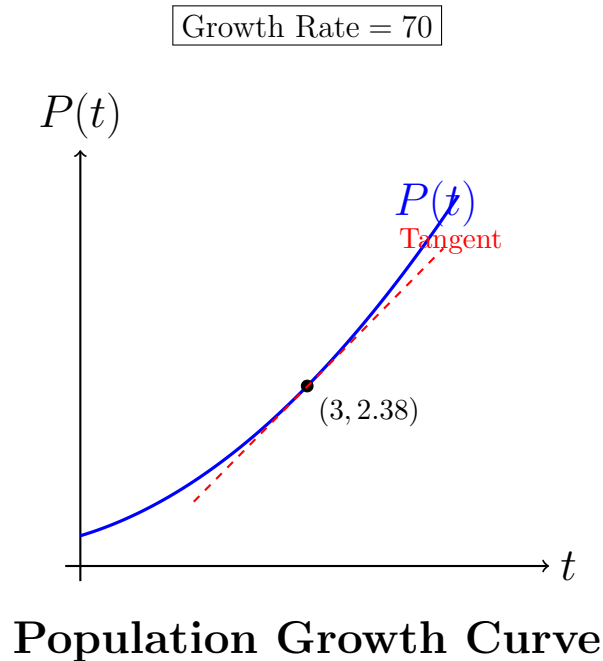
Substitute:

$$t = 5$$

$$\frac{dP}{dt} = 50 + 4(5)$$

$$= 50 + 20$$

$$= 70$$



10.3 Average Cost

Average cost represents cost per unit produced.

If total cost function is:

$$C(x)$$

then average cost is:

$$AC = \frac{C(x)}{x}$$

Average Cost Formula

$$AC = \frac{C(x)}{x}$$

Solved Example

Suppose:

$$C(x) = x^2 + 100$$

Find the average cost.

Solution

$$\begin{aligned} AC &= \frac{x^2 + 100}{x} \\ &= x + \frac{100}{x} \end{aligned}$$

$$AC = x + \frac{100}{x}$$

10.4 Marginal Cost

Marginal cost is the additional cost of producing one extra unit.

If:

$$C(x)$$

is total cost function, then:

$$MC = \frac{dC}{dx}$$

Marginal Cost Formula

$$MC = \frac{dC}{dx}$$

Solved Example

Suppose:

$$C(x) = x^3 + 5x + 10$$

Find marginal cost.

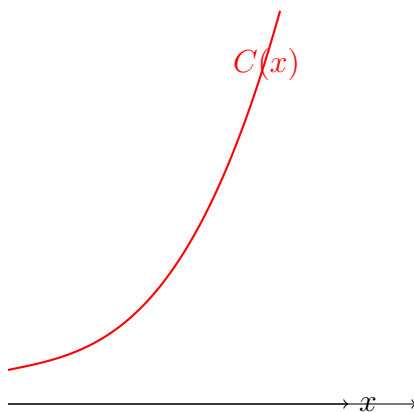
Solution

Differentiate:

$$MC = \frac{dC}{dx}$$

$$= 3x^2 + 5$$

$$MC = 3x^2 + 5$$



Cost Function

10.5 Marginal Revenue

Marginal revenue measures additional revenue earned by selling one extra unit.

If:

$$R(x)$$

is the revenue function, then:

$$MR = \frac{dR}{dx}$$

Marginal Revenue Formula

$$MR = \frac{dR}{dx}$$

Solved Example

Suppose:

$$R(x) = 50x - x^2$$

Find marginal revenue.

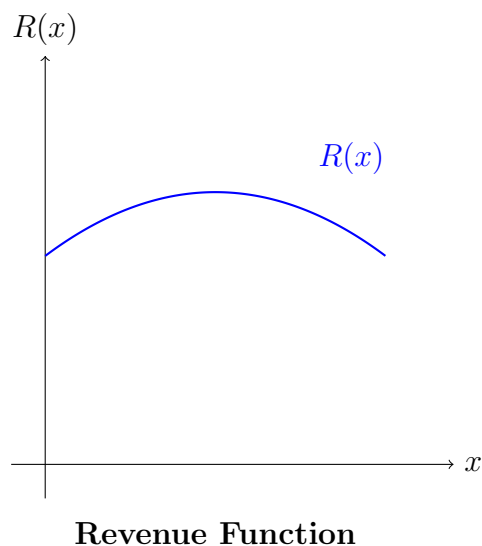
Solution

Differentiate:

$$MR = \frac{dR}{dx}$$

$$= 50 - 2x$$

$$MR = 50 - 2x$$



10.6 Summary Table

Concept	Formula
Growth Rate	$\frac{dy}{dt}$
Average Cost	$\frac{C(x)}{x}$
Marginal Cost	$\frac{dC}{dx}$
Marginal Revenue	$\frac{dR}{dx}$

Important Notes

- Differentiation helps analyze real-life problems.
- Marginal values measure instantaneous change.
- Growth rate shows how quickly quantities increase or decrease.
- Businesses use derivatives to maximize profit and minimize cost.

10.7 Solved Questions

Question 1: Growth Rate

If:

$$P(t) = 500 + 30t + 2t^2$$

find the growth rate at:

$$t = 4$$

Solution

Differentiate:

$$\frac{dP}{dt} = 30 + 4t$$

At:

$$t = 4$$

$$\frac{dP}{dt} = 30 + 16$$

$$= 46$$

Growth Rate = 46

Question 2: Average Cost

If:

$$C(x) = x^2 + 200$$

find the average cost.

Solution

$$AC = \frac{C(x)}{x}$$

$$= \frac{x^2 + 200}{x}$$

$$= x + \frac{200}{x}$$

$$\boxed{AC = x + \frac{200}{x}}$$

Question 3: Marginal Cost

If:

$$C(x) = 2x^3 + 5x + 10$$

find marginal cost.

Solution

$$MC = \frac{dC}{dx}$$

$$= 6x^2 + 5$$

$$\boxed{MC = 6x^2 + 5}$$

Question 4: Marginal Revenue

If:

$$R(x) = 60x - x^2$$

find marginal revenue.

Solution

$$MR = \frac{dR}{dx}$$

$$= 60 - 2x$$

$$\boxed{MR = 60 - 2x}$$

Question 5

The population of a town is given by:

$$P(t) = 1000 + 40t + 3t^2$$

Find the growth rate when:

$$t = 5$$

Solution

Growth rate is:

$$\frac{dP}{dt}$$

Differentiate:

$$\frac{dP}{dt} = 40 + 6t$$

At:

$$t = 5$$

$$\frac{dP}{dt} = 40 + 6(5)$$

$$= 40 + 30$$

$$= 70$$

Growth Rate = 70

Question 6

The total cost function is:

$$C(x) = x^2 + 500$$

Find the average cost.

Solution

Average cost is:

$$AC = \frac{C(x)}{x}$$

Substitute:

$$AC = \frac{x^2 + 500}{x}$$

Simplify:

$$AC = x + \frac{500}{x}$$

$$AC = x + \frac{500}{x}$$

Question 7

The total cost function is:

$$C(x) = 3x^2 + 10x + 100$$

Find the marginal cost.

Solution

Marginal cost is:

$$MC = \frac{dC}{dx}$$

Differentiate:

$$MC = \frac{d}{dx}(3x^2 + 10x + 100)$$

$$= 6x + 10$$

$$MC = 6x + 10$$

Question 8

The revenue function is:

$$R(x) = 80x - x^2$$

Find the marginal revenue.

Solution

Marginal revenue is:

$$MR = \frac{dR}{dx}$$

Differentiate:

$$MR = \frac{d}{dx}(80x - x^2)$$

$$= 80 - 2x$$

$$MR = 80 - 2x$$

Question 9

The cost function of a company is:

$$C(x) = x^3 + 4x^2 + 10x + 50$$

Find:

1. Average Cost
2. Marginal Cost

Solution**(i) Average Cost**

$$\begin{aligned} AC &= \frac{C(x)}{x} \\ &= \frac{x^3 + 4x^2 + 10x + 50}{x} \\ &= x^2 + 4x + 10 + \frac{50}{x} \end{aligned}$$

$$\boxed{AC = x^2 + 4x + 10 + \frac{50}{x}}$$

(ii) Marginal Cost

$$MC = \frac{dC}{dx}$$

Differentiate:

$$MC = 3x^2 + 8x + 10$$

$$\boxed{MC = 3x^2 + 8x + 10}$$

10.8 Practice Questions

1. If:

$$P(t) = 500 + 20t + t^2$$

find growth rate at:

$$t = 4$$

2. If:

$$C(x) = x^2 + 200$$

find average cost.

3. Find marginal cost if:

$$C(x) = 2x^3 + 5x + 1$$

4. Find marginal revenue if:

$$R(x) = 100x - x^2$$

5. If:

$$R(x) = 40x - 2x^2$$

find:

$$MR$$

Euclid

“There is no royal road to geometry.”

Mathematics requires effort and cannot be shortcut.